



CONTACT

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EDUCATION

Intermediate
Fsc Pre-Engineering
KRL Model College For Boys, Kahuta
2018 — 2020

BS Computer Science
Institute of Space Technology (IST),
Islamabad
2022 — 2026 (ongoing)

CORE SKILLS

- ▶ Java · Python · C++ · JavaScript
- ▶ React & Node.js · FastAPI
- ▶ HTML5 & CSS3 · Vercel
- ▶ MongoDB · MySQL · Firebase Firestore
- ▶ AI/ML Model Training & Testing
- ▶ Mobile App Development
- ▶ Creo Parametric

ACHIEVEMENT

HEC National Skill Competency Test (NSCT)
95.3 Percentile
HEC-conducted CS competency assessment

ACHIEVEMENT

- ▶ Certificate of Internship — R&D and Electronics Systems Support (RESS) Lab
- ▶ Arduino Workshop-IEEE IST Student Branch Certificate of Participation
- ▶ COMPEC 2025,NUST College of EME Speed programming competition

SYED MAZHAR HUSSAIN SHAH
CYBERSECURITY & FULL-STACK SOFTWARE DEVELOPER

PROFESSIONAL SUMMARY

Computer Science undergraduate at the Institute of Space Technology with hands on experience across cybersecurity, AI/ML, and full-stack development . As a passionate learner and tech enthusiast, I excel at turning complex challenges into innovative solutions. I bring a unique blend of technical and creative skills to the table.

FINAL YEAR PROJECT

Intelligent Extended Detection & Response (XDR) Platform

Cybersecurity · AI/ML

- Architected four independent AI-driven detection engines — system behavior, user behavior, network activity, and malware analysis — each producing an independent threat score.
- Built a fusion engine that combines the four engine scores into a single composite risk score used to classify activity as malicious or benign.
- Implemented automated decision logic that triggers appropriate mitigation actions based on the fused threat score, closing the loop from detection to response.

RESEARCH EXPERIENCE

Research Intern — R&D and Electronics Systems Support (RESS) Lab

Dr. A.Q. Khan Research Laboratories, Kahuta · 2-Month Internship

- Studied the architecture of GRBLHAL, an open-source CNC motion-control firmware for 32-bit microcontrollers, including G-code parsing, motion planning, and its hardware abstraction layer.
- Connected firmware concepts to real industrial CNC hardware and legacy retrofit control systems, and documented findings for the engineering team.

PROJECTS

Smart Retail System — Real-Time Product Detection & Billing

Python · YOLOv8 · OpenCV · Flask

- Built a computer vision system that detects retail products in real time and automatically generates a shopping cart and bill.
- Independently collected and annotated the dataset across varied lighting, angles, and backgrounds using Label Studio, then iterated through several rounds of relabeling and retraining to resolve misclassification issues.
- Implemented a temporal smoothing buffer to eliminate flickering detections and improve real-time stability across live webcam, image, and video input.

DripBlood — Emergency Blood Donor Network

Android · Firebase · Google Maps API

- Developed a full-stack Android application connecting people in urgent need of blood with nearby donors and donation camps via a live, range-based map.
- Implemented multi-method authentication (email/password, Google, Facebook, OTP), a voice-and-text medical assistant chatbot, and an admin dashboard for managing users, camps, and live statistics.
- Built on Firebase Auth, Realtime Database, and Storage, with Cloudinary handling media uploads to control costs.